

Part I

SUMMARY of Derivatives as Functions

Summary

February 21, 2026

Summary

- The derivative of a function f is a function f' whose domain consists of all points in the domain of f where the function f is differentiable. The values of f' are given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

where this value exists.

- If f is differentiable at $x = a$, then f is continuous at $x = a$.

Part II

Recitation Questions

Problem 1

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Problem 1 (a) Suppose $f'(2)$ exists. Which of the following must be true?

- (i) $\lim_{x \rightarrow 2} f(x)$ must exist, but $\lim_{x \rightarrow 2} f(x) \neq f(2)$
- (ii) $\lim_{x \rightarrow 2} f(x) = f(2)$.
- (iii) $\lim_{x \rightarrow 2} f(x) = f'(2)$
- (iv) $\lim_{x \rightarrow 2} f(x)$ need not exist.

Explanation. The correct answer is (ii):

$$\begin{aligned} f'(2) \text{ exists} &\iff f \text{ differentiable at } x = 2 \\ &\implies f \text{ continuous at } x = 2 \\ &\iff \lim_{x \rightarrow 2} f(x) = f(2) \end{aligned}$$

Assuming that $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$, which of the following is true?

- (b) (i) $\frac{0}{0} = 1$
- (ii) the tangent line to $y = \sin(x)$ at $(0, 0)$ has slope 1.
- (iii) you can cancel the x 's.
- (iv) for all x near 0, $\sin(x) = x$.
- (v) for all x near 0, $\sin(x) \approx x$.

Explanation. This problem has two correct answers: (ii) and (v).

Statement (ii) is true:

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{\sin(0+h) - \sin(0)}{h} \text{ Which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \\ &= 1 \\ &\implies \text{slope of tangent line at } (0, 0) \text{ is } 1 \end{aligned}$$

Statement (v) is true: When x is near 0 the tangent line $y = x$ is a good approximation to f .

Problem 2

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Problem 2

(a) Fill in the blanks

$$f'(x) = \lim_{h \rightarrow 0} \frac{???}{h}$$

if the limit exists.

Explanation.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

if the limit exists.

(b) Let

$$f(x) = \frac{1}{x+4}.$$

Use the (limit) definition of derivative in (a) to find $f'(x)$.

Explanation.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h+4} - \frac{1}{x+4}}{h} \text{ which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0} \frac{\frac{x+4 - (x+h+4)}{(x+h+4)(x+4)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{-h}{(x+h+4)(x+4)} \\ &= \lim_{h \rightarrow 0} \frac{-1}{(x+h+4)(x+4)} = \frac{-1}{(x+4)^2} \end{aligned}$$

[3]

Problem 3

February 21, 2026

Problem 3 Let $f(x) = |5 - x|$.

(a) For $a < 5$, find $f'(a)$.

Explanation. Recall that

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

When $a < 5$, $0 < 5 - a$ and so $f(a) = |5 - a| = 5 - a$. Since we are considering the limit as x approaches a , (and therefore interested in values of x really close to a), we may also assume that $x < 5$ and therefore $f(x) = 5 - x$. Thus

$$\begin{aligned} f'(a) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{(5 - x) - (5 - a)}{x - a} \text{ which has form } \frac{0}{0} \\ &= \lim_{x \rightarrow a} \frac{-(x - a)}{x - a} \\ &= \lim_{x \rightarrow a} -1 = -1. \end{aligned}$$

(b) For $a > 5$, find $f'(a)$.

Explanation. When $a > 5$, $0 > 5 - a$ and so $f(a) = |5 - a| = -(5 - a) = a - 5$. Since we are considering the limit as x approaches a , we may also assume that $x > 5$ and therefore $f(x) = x - 5$. Thus

$$\begin{aligned} f'(a) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{(x - 5) - (a - 5)}{x - a} \text{ which has form } \frac{0}{0} \\ &= \lim_{x \rightarrow a} \frac{x - a}{x - a} \\ &= \lim_{x \rightarrow a} 1 = 1. \end{aligned}$$

(c) Determine whether $f'(5)$ exists.

Explanation. If $f'(5)$ exists then $\lim_{x \rightarrow 5} \frac{f(x) - f(5)}{x - 5}$ exists. But when x is close to 5 it may be that $x > 5$ or $x < 5$. So in order to find this limit, we have to check whether the two one-sided limits are equal.

$$\text{This means, } \lim_{x \rightarrow 5^-} \frac{f(x) - f(5)}{x - 5} = \lim_{x \rightarrow 5^+} \frac{f(x) - f(5)}{x - 5}$$

Each of these has form $\frac{0}{0}$.

Problem 3

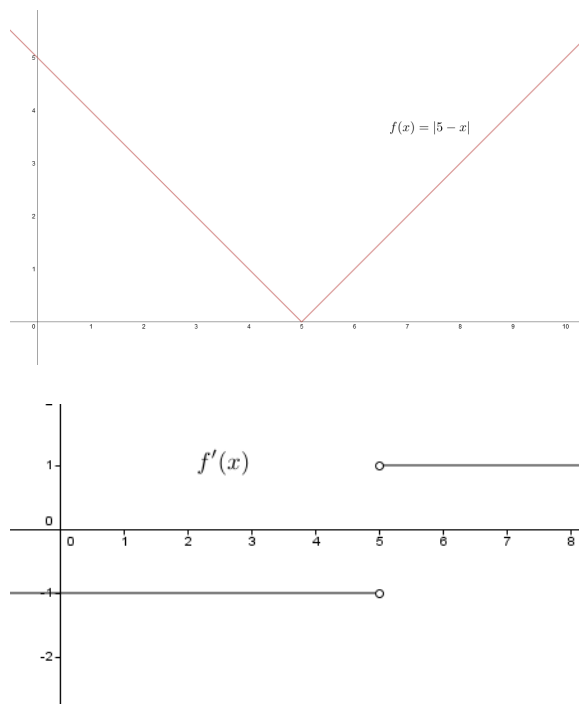
$$\begin{aligned}\lim_{x \rightarrow 5^-} \frac{f(x) - f(5)}{x - 5} &= \lim_{x \rightarrow 5^-} \frac{-(x - 5) - (0)}{x - 5} = -1 \\ \lim_{x \rightarrow 5^+} \frac{f(x) - f(5)}{x - 5} &= \lim_{x \rightarrow 5^+} \frac{(x - 5) - (0)}{x - 5} = 1 \\ \lim_{x \rightarrow 5^-} \frac{f(x) - f(5)}{x - 5} &\neq \lim_{x \rightarrow 5^+} \frac{f(x) - f(5)}{x - 5}\end{aligned}$$

Therefore, $f'(5)$ does not exist.

$$f'(a) = \begin{cases} 1 & \text{if } a > 5 \\ \text{undefined} & \text{if } a = 5 \\ -1 & \text{if } a < 5 \end{cases}$$

(d) Sketch a graph of the function $f(x)$ and its derivative $f'(x)$

Explanation.

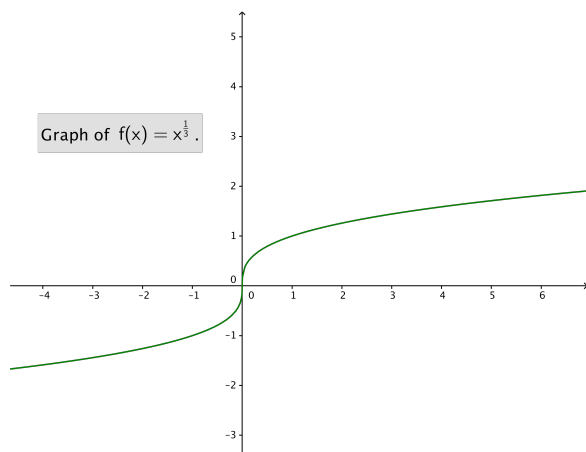


Problem 4

February 21, 2026

Problem 4

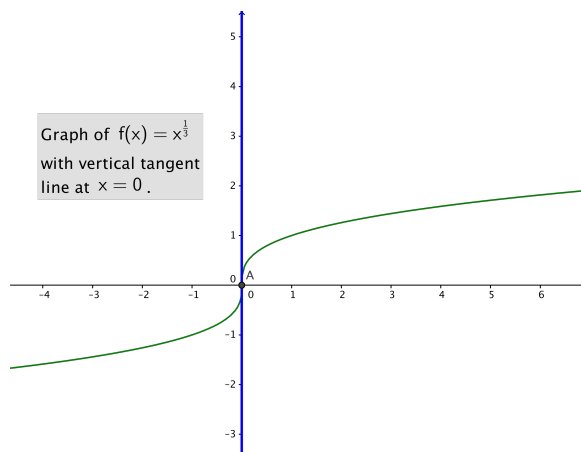
Problem 4 Define the function f by $f(x) = x^{1/3}$ and consider the graph of this function:



Which of the following two statements are true:

- (a) The graph of f has a tangent line at $x = 0$.

Explanation. This statement is **true!** The function f has a vertical tangent at $x = 0$:



- (b) The derivative $f'(0)$ is defined.

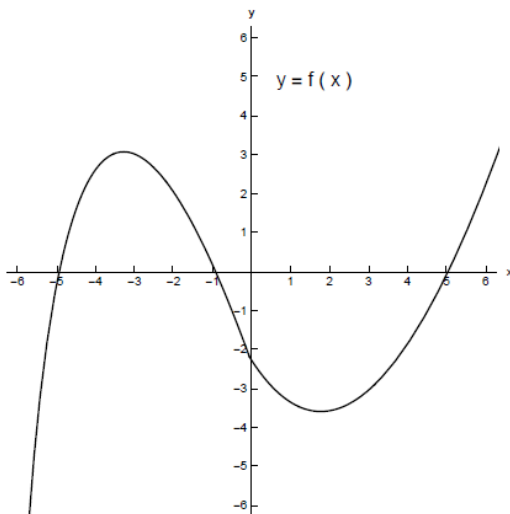
Explanation. This statement is **false**!

$$\begin{aligned}\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{h^{1/3}}{h} \text{ which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0^+} \frac{1}{h^{\frac{2}{3}}} \\ &= \infty \text{ because the limit is of the form: } \frac{\text{pos}}{0^+} \\ &\implies f'(0) \text{ is undefined}\end{aligned}$$

Problem 5

February 21, 2026

Problem 5 Suppose we are given the graph of a function f :



- (a) Use this graph to find the following: (Assume all values will be integers or $+\infty$ or $-\infty$)

- (i) all x where $f(x) = 0$,

Explanation. $f(x)$ is zero when the function crosses the x -axis. Therefore $f(x) = 0$ when $x = -5$, $x = -1$, and $x = 5$.

- (ii) all x where $f(x) > 0$,

Explanation. $f(x)$ is positive when the graph of the function is above the x -axis. Therefore $f(x) > 0$ on $(-5, -1) \cup (5, \infty)$.

- (iii) all x where $f(x) < 0$, and

Explanation. $f(x)$ is negative when the graph of the function is below the x -axis. Therefore $f(x) < 0$ on $(-\infty, -5) \cup (-1, 5)$.

- (iv) all x where $f(x)$ attains a local maximum and all x where f attains a local minimum.

Explanation. $f(x)$ has a local maximum at $x = -3$. $f(x)$ has a local minimum at $x = 2$.

Without sketching the graph of f' find

- (b) (i) all x where $f'(x) = 0$,

Explanation. $f'(x)$ is zero when the tangent line has a slope of zero, which is approximately at $x = -3$ and $x = 2$. Note, for this question, these are the same answers as the (local) highest and lowest point for the graph of f .

- (ii) all x where $f'(x) > 0$,

Explanation. $f'(x)$ is positive when the slope of the tangent line is positive. Observe that f is increasing on $(-\infty, -3), (2, \infty)$ and this same set of intervals is where the tangent lines have positive slope. Therefore $f'(x) > 0$ on $(-\infty, -3), (2, \infty)$.

- (iii) all x where $f'(x) < 0$, and

Explanation. $f'(x)$ is negative when the slope of the tangent line is negative. Observe that f is decreasing on $(-3, 2)$ and this interval is where the tangent lines have negative slope. Therefore $f'(x) < 0$ on $(-3, 2)$.

- (iv) On the following intervals, does $f'(x)$ seem to be increasing or decreasing?

- i. $(-\infty, 0)$

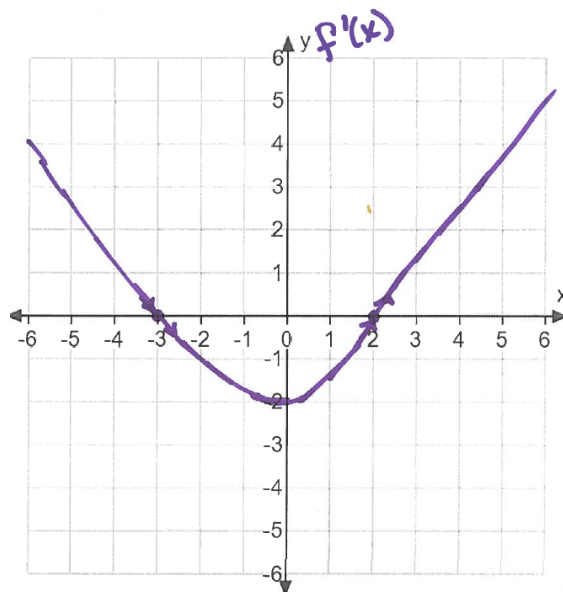
Explanation. decreasing

- ii. $(0, \infty)$

Explanation. increasing

- (c) Sketch a graph of f' .

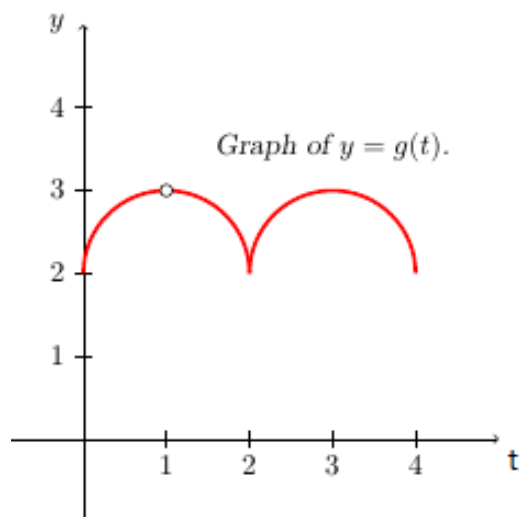
Explanation. The graph of f' is approximately



Problem 6

February 21, 2026

Problem 6 Use the graph of g



- (a) Find the values of t in $(0, 4)$ at which g is not continuous.

Explanation. g is not continuous at $t = 1$.

- (b) Find the values of t in $(0, 4)$ at which g is not differentiable.

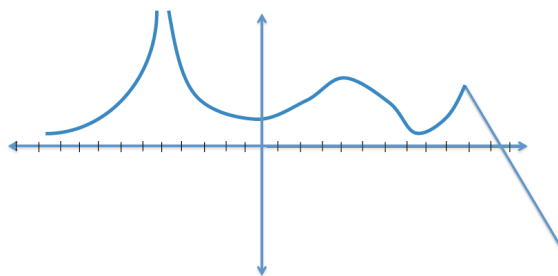
Explanation. g is not differentiable at $t = 1$, because g is not continuous there, and at $t = 2$, because g has a "cusp" there.

Problem 7

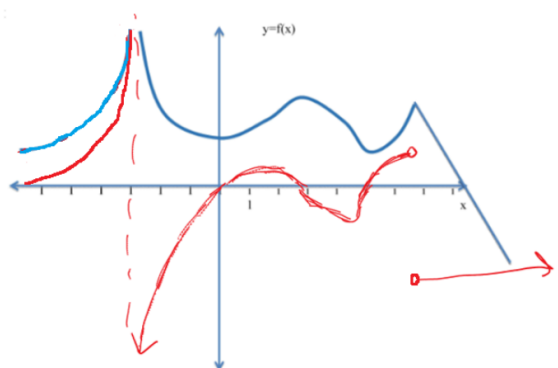
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Problem 7

Problem 7 Given the following graph of a function h sketch a graph of the derivative h' .



Explanation. The graph of the derivative is in red. **Important Note:** Despite being drawn on the same graph, the “units” for f and f' are *not* the same!



[2]